Decentralized Digital Cancer Screening Network for Densely Populated Urban Areas (DigiNode)

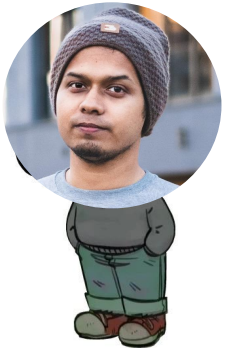
Digital Health Systems

Group-5

Who we are

Md Nahian Imtiaz Hasan

B.Sc. in IT



Raihanul Hafiz Sourav

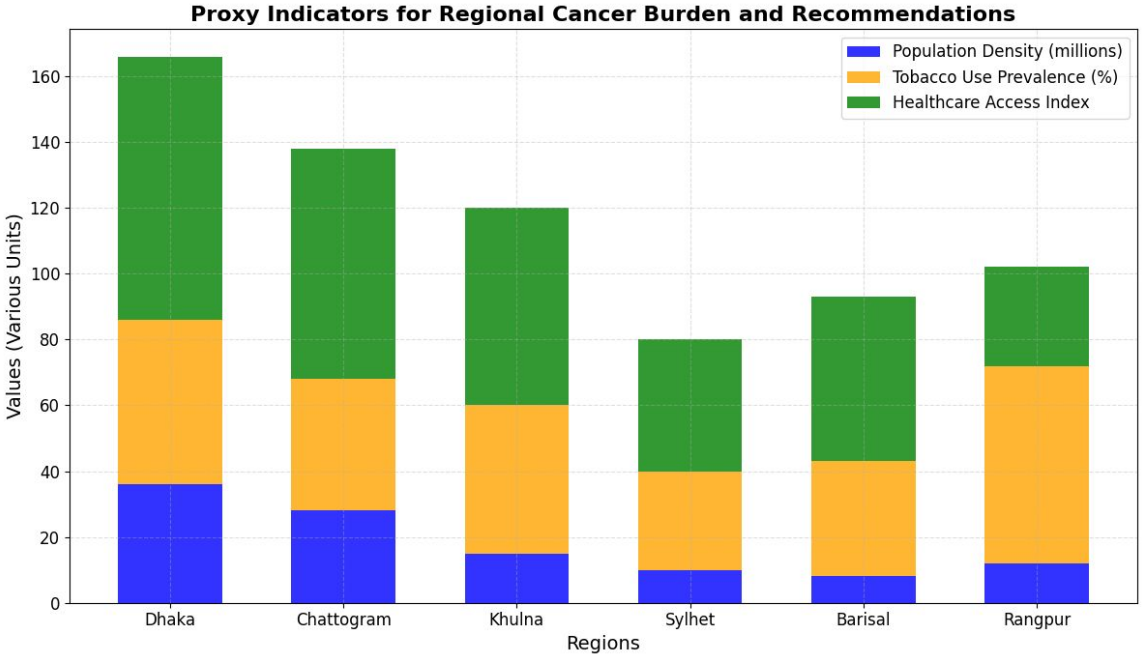
B.Sc. in CSE

Introduction to the Problem

- Only 4 public cancer hospitals
- All in Dhaka
- National Institute of Cancer Research & Hospital (NICRH)
- Long distance travel
- Waiting period

Target Region

- **Bangladesh had approximately 167,256 new cancer cases**
- **Age-standardized incidence rate of 105.6 per 100,000 individuals**
- **Detailed regional cancer incidence rates are not readily available**



Source: Global Cancer Observatory's 2022

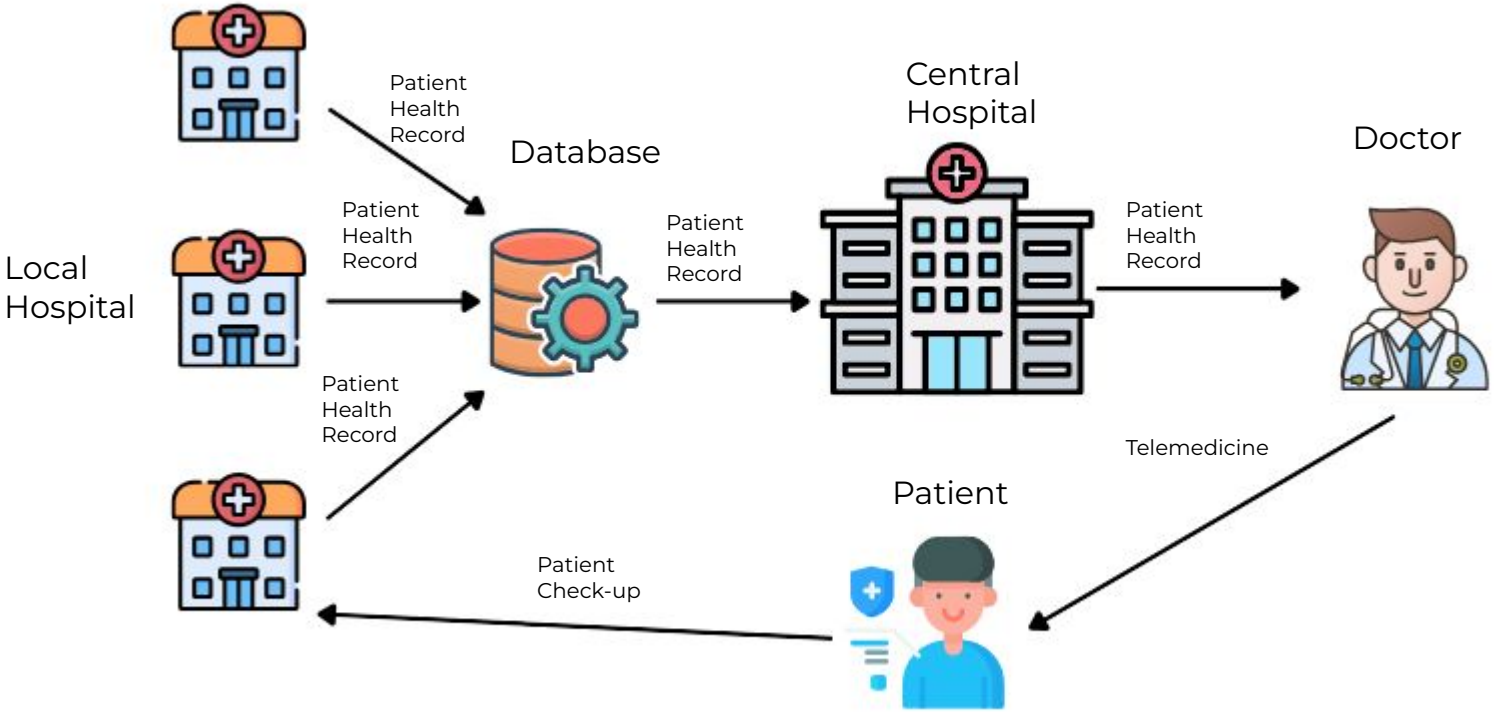
Goal

Bridging geographical barriers in cancer screening

Value Proposal

Reducing Cancer mortality by enhancing screening accessibility

Our Solution



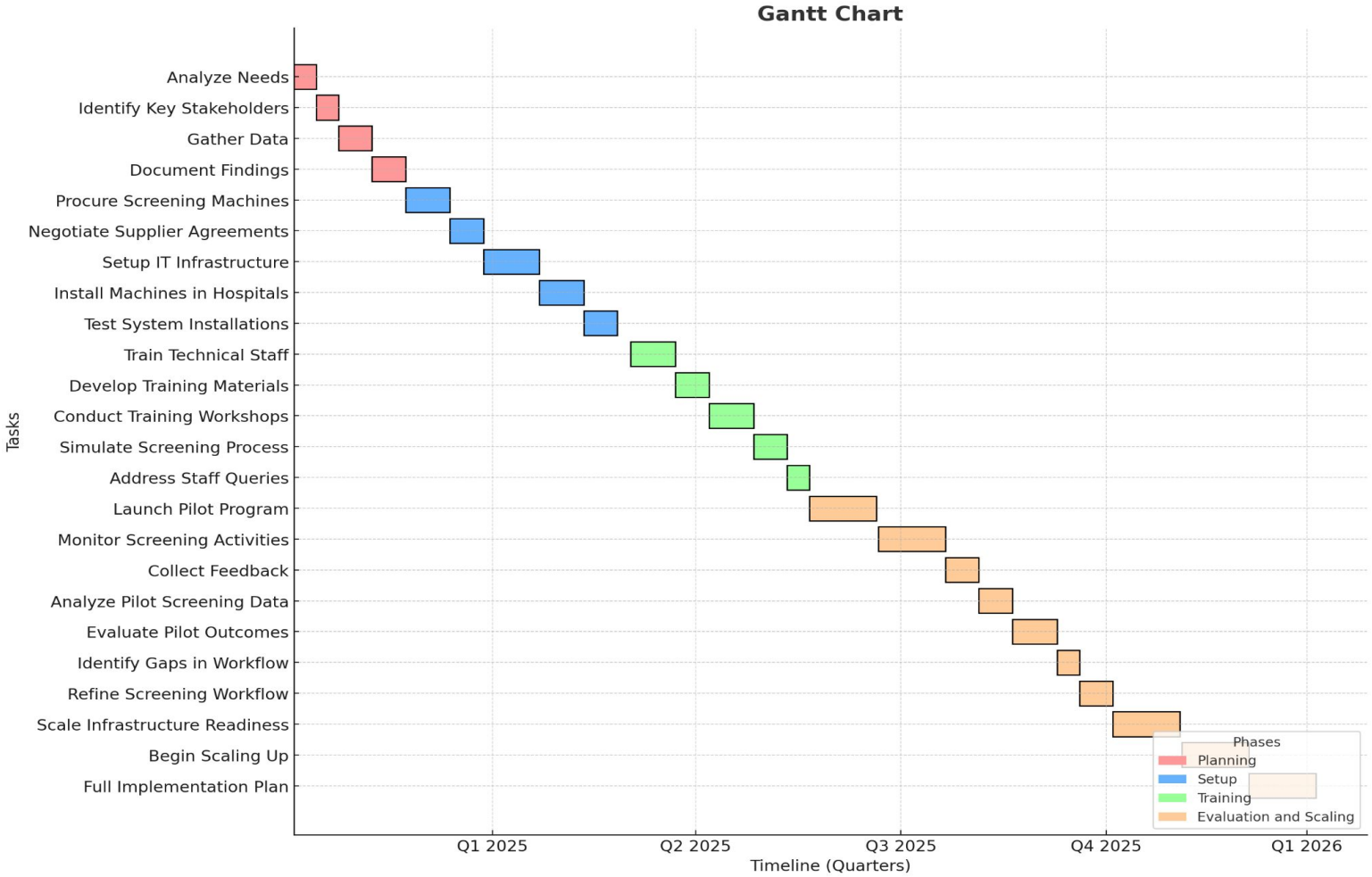
Value Chain

Data Sharing Platform	IT systems for data sharing, machine installation at local hospitals.
Workforce Development	Training programs for Employees.
Integrating AI	Applying latest AI Algorithms in Analysis
Supply Chain	Ensuring reliable supply chain

Resource Management	**Screening Operations**	**Case Forwarding**	**Community Outreach**	**Follow up Services**
Setting up screening infrastructure in regional hospitals	Conducting cancer screening and Data Collection	Transferring critical patient data records to specialized cancer hospitals	Promoting screening awareness through local campaigns	Follow up care and personalized support

Value for Patients

Project Gantt Chart



Project Milestones

Planning and Assessment

Analyzing needs

January-February

1

Infrastructure Setup

Procurement and Installment

March-May

2

Training and Pilot Program

Training staff and launching project

June- September

3

Evaluation and Scaling

Analyze pilot outcomes,refine and scale up

October-December

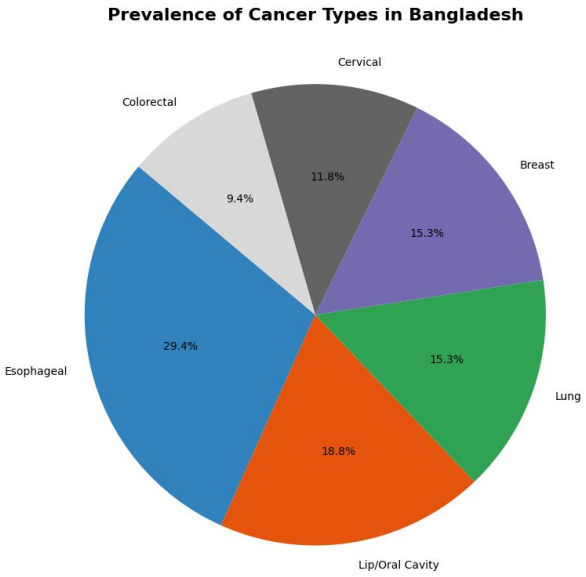
4

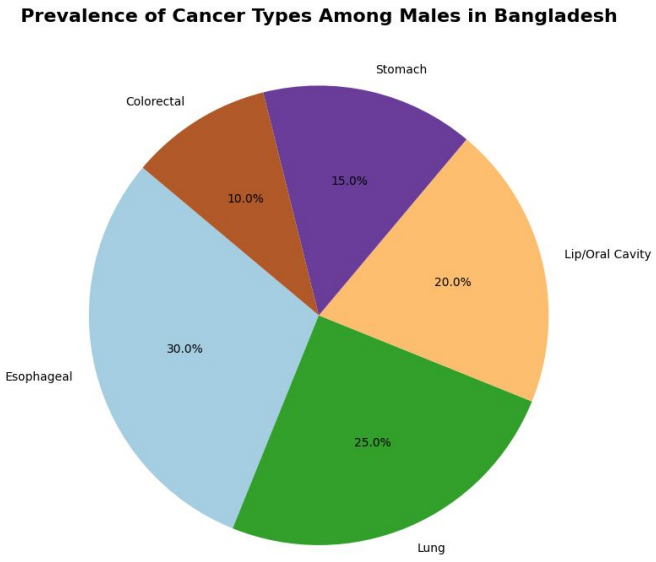
2025

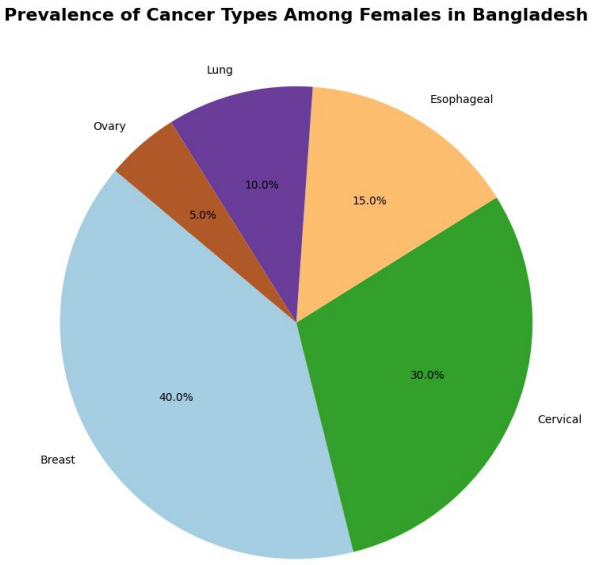
Business Model

Customer Segments (For Whom?)	Key Activities (What Do We Do?)	Key Resources (Whom Do We Work With?)
Primary Customers: Rural and underserved people. **Stakeholders:** • Government Hospitals and Healthcare System(Policy and Funding) • NGOs and healthcare organizations (advocacy and grants). • Private clinics for future scalability.	**Screening Programs**: Localized cancer detection initiatives. **Infrastructure Setup**: Deploy machines, IT systems, and AI tools. **Training**: Equip healthcare workers with skills to run programs. **Monitoring & Scaling**: Evaluate pilots and expand successful models.	• Government hospitals for policy support and infrastructure sharing • NGOs and international organizations for funding, advocacy, and campaigns. • Private sector (equipment suppliers, IT service providers). • Local community leaders to drive awareness and adoption
Value Proposition (Why us?)	**Cost Structure (What Do We Spend On?)**	**Revenue Streams (How do we earn?)**
• **Accessible and affordable:** Accessible and affordable cancer diagnostics for underserved rural populations. • **Early Detection:** Early Detection to improve survival rate and reduce treatment costs. • **Empowerment:** Empower smaller hospitals with modern equipment and expertise.	**Fixed Cost:** Screening Equipment IT Infrastructure **Variable Expense:** Personnel Training and Salaries Operations & Maintenance **Contingency Plans:** Reserve fund **Scalability Plan:** Expand to additional districts over 3-5 years based on pilot success.	• **Pay-Per-Screening**: Minimal fee per screening. • **Grants & Funding**: Partnerships with governments and health organizations. • **Research Licensing**: Long-term opportunity for anonymized data use.

Target Cancer Types







Estimated Cost for Screening Equipments

Equipment	Purpose	Quantity	Unit Price (EUR)	Total Cost (EUR)
Digital Mammography Machine	Breast Cancer Screening	2	€250,000	€500,000
Ultrasound Machine	Tumor Detection	4	€80,000	€320,000
CT Scanner	Advanced Imaging	2	€800,000	€1,600,000
MRI Scanner	Soft Tissue Evaluation	1	€2,000,000	€2,000,000
Colposcope	Cervical Cancer Detection	3	€15,000	€45,000
HPV Testing Equipment	Cervical Cancer Risk Assessment	2	€40,000	€80,000
Endoscopy Equipment	Colorectal and GI Cancer Detection	2	€90,000	€180,000

Estimated Cost for Data Center

Component	Purpose	Quantity	Unit Price (EUR)	Total Cost (EUR)
Cloud Storage Servers	Store EHR and imaging data securely	1	€15,000	€15,000
Server Software Licenses	Cloud management, encryption, and access	1	€5,000	€5,000
Telecommunication Setup	High-speed connectivity for data transfer	2	€10,000	€20,000
Data Backup Systems	Redundant systems to prevent data loss	1	€10,000	€10,000
Workstation Computers	Local hospital terminals for EHR access	5	€2,000	€10,000
IT Security Solutions	Firewalls, encryption, and data protection	1	€8,000	€8,000
Software Development and Integration	EHR software customization and installation	1	€15,000	€15,000
Training for IT Staff	Training for system usage and maintenance	1	€3,000	€3,000

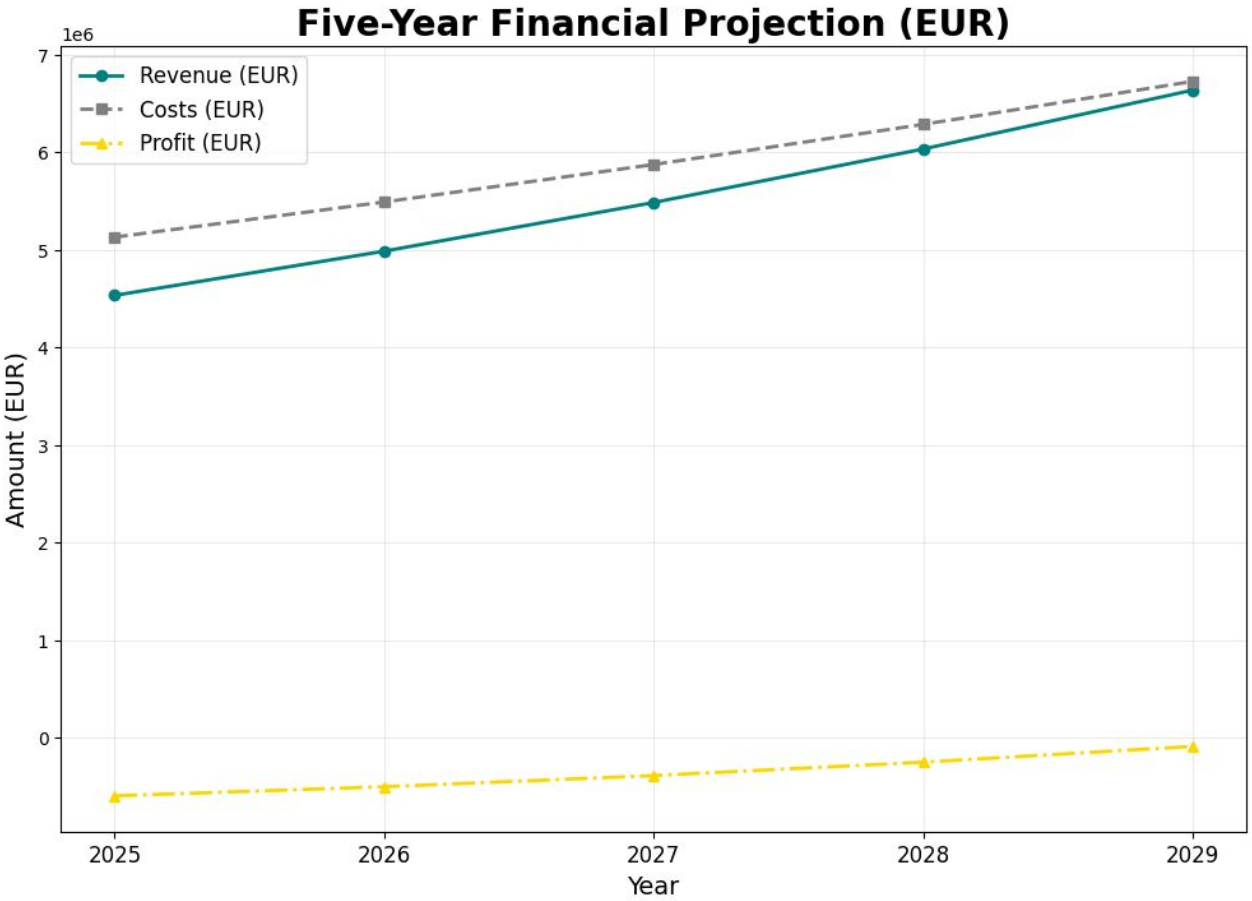
First Year Financial Statement

Category	Item	Amount
Fixed Cost	Screening Equipment	€4,725,000
	Data Center	€86,000
	Software	€30,000
Variable Expense	Maintenance	€10,000
	Medical Supplies	€50,000
	Operation	€12,000
	Personnel	€168,000
Contingency Plan	Reserve Fund	€50,000
Total		€5,131,000

Five Year Financial Projection

Year	Revenue	Costs	Profit
2025	€4,533,727	€5,131,000	-€597,273
2026	€4,987,100	€5,491,170	-€504,070
2027	€5,485,810	€5,875,551	-€389,741
2028	€6,034,391	€6,286,840	-€252,449
2029	€6,637,830	€6,727,919	-€90,088

Five Year Financial Projection



Recommendation

- **Start Small**: Begin with key equipments and run a pilot program.
- **Find Funding**: Focus on grants, government support, and health organizations. Hiring a dedicated finance team can balance the budgets.
- **Measure Progress**: Track results like screenings completed and detection rates.

Conclusion

- The project is realistic and well-planned with clear costs and goals.
- Early cancer detection will save lives, especially in underserved areas.
- The model can grow over time with the right technology and funding.

Thanks!